

Independent Test Solution



Evidence decision

Technology: Java 26, Maven, JUnit 5, Cucumber, Jackson, JSON Schema, AssertJ

+

TECHNOLOGY STACK

Deterministic services, readable evidence.

Java 26 Core validation and reporting services.

Maven Build, dependency, and repeatable execution.

JUnit 5 Unit, mutation, and contract regression tests.

Cucumber Segment-wise business-readable workflows.

Jackson JSON parsing and canonical model mapping.

JSON Schema Structural contract validation.

AssertJ Readable assertions and evidence checks.

PowerShell/Python ATL105 ingestion, comparison, and report automation.

+

REPOSITORY ARCHITECTURE

Reusable engine, specification-specific knowledge.

Reusable Java engine

src/main/java

→

ATL105 specification pack

rules, schemas, scripts

→

Evidence and reports

test-output

regression-artifact-validator/

src/main/java/com/coreauth/validator/

canonical/ coverage/ paths/ rules/ source/ traceable/ validation/

src/test/java/ # unit, contract, mutation, Cucumber
tests

specifications/ATL105/

contract/ docs/ reports/ schemas/ scripts/

test-input/ test-output/

pom.xml

A new specification supplies a new pack; reusable validation services remain stable.

+

SPECIFICATION PACK

ATL105 is the independent knowledge boundary.

Source knowledge

Extracted ATL105 specification, source sections, page references, code tables, and canonical anchors.

Segment knowledge

Field rules, applicability, companion segments, lifecycle behavior, serialization, and specialized domains.

Execution contract

Schemas, Test Solution packages, synthetic fixtures, coverage profiles, and validation status.

Source evidence

→

Rule catalog

→

Canonical anchors

→

Validation oracle

+

FRAMEWORK LAYERS

Five technical layers make the decision auditable.

Canonical model CanonicalArtifactPackage, CanonicalRequirement, CanonicalScenario, CanonicalTestCase, CanonicalTestData, PackageManifest, SourceAnchor.

Rule oracle ATL105 segment catalogs with rule IDs, elements, source sections, conditions, severity, evidence, and review status.

Validation services Schema, provenance, semantic, compatibility, lifecycle, serialization, and mutation validation.

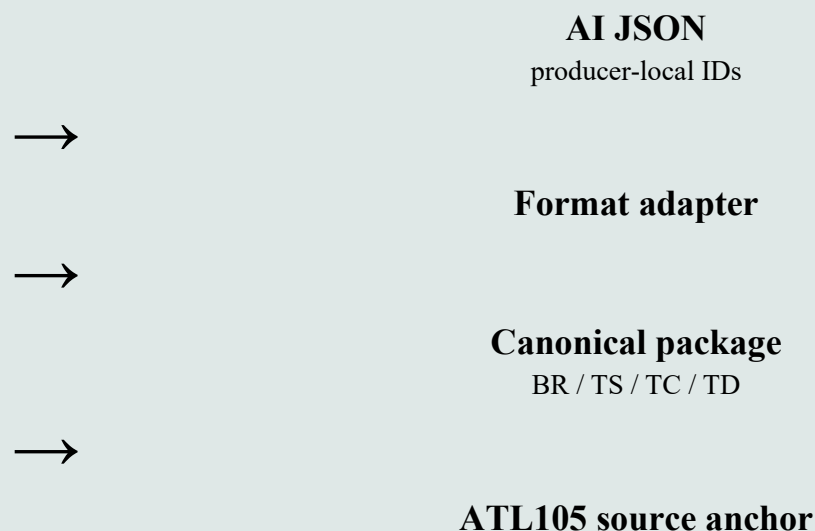
Evidence services Traceability matrices, four-level matching, gap reports, coverage ratios, and review queues.

Governance Approval states, SME/TBA decisions, manual sample gates, release evidence, and Fiserv acceptance.

+

CANONICAL MODEL

Different producer IDs, one semantic identity.



Preserve

Original producer IDs, files, metadata, source version, and hashes remain available.

Normalize

Different AI/Test formats map to one neutral artifact contract.

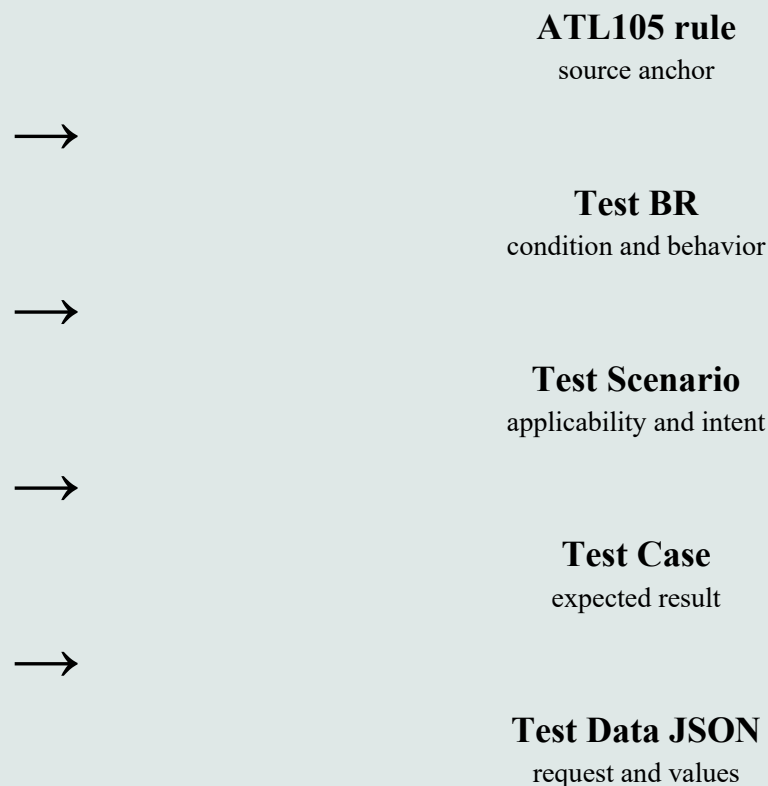
Compare

Matching uses canonical anchors and business meaning, not text similarity alone.

+

VALIDATION CHAIN

The framework proves the complete artifact chain.



Graph checks

IDs, links, orphans, duplicates, and anchor continuity.

Semantic checks

Fields, values, conditions, lifecycle, and companion rules.

Readiness checks

Payload, expected outcome, serialization, and execution status.

+

VALIDATION SERVICES

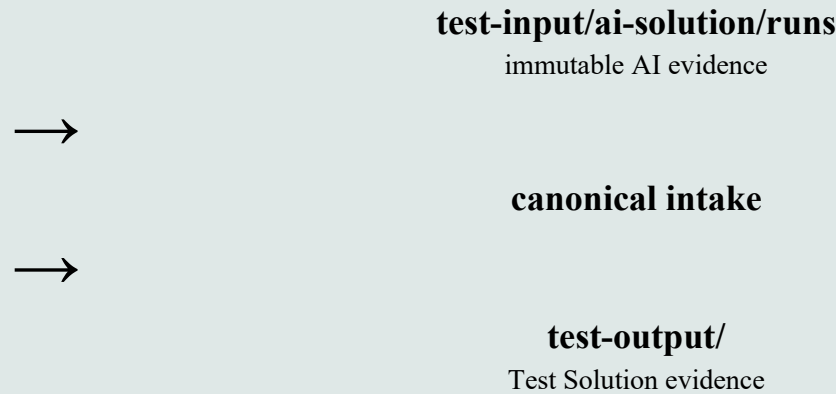
Each validator owns a specific risk.

Schema and JSON	Parseability, object structure, required fields, contract shape	Schema findings and package status
Source and provenance	Specification version, source anchors, package identity, hashes	Provenance decision
Segment validators	Identity, fields, values, lengths, applicability, compatibility	Rule-level validation errors
Lifecycle and serialization	Sequence, retries, reversals, separators, lengths, wire format	Execution and mutation evidence
Traceability and coverage	BR → TS → TC → TD graph and independent metrics	Coverage and review reports

+

STORAGE AND OWNERSHIP

Inputs, baselines, and decisions are separated.



Test baselines

test-brs/, test-scenarios/, test-cases/, and test-json/.

Review evidence

traceability-matrix/, ai-solution-independent-review/, and gap reports.

Coverage reports

Segment-specific reports under ai-artifacts/coverage-reports/segment-`<id>`/.
+

COVERAGE AND MUTATION

Coverage is calculated; validator strength is tested.

Rule Mandatory ATL105 rules with executable evidence.

Chain BRs connected through TS, TC, and TD JSON.

Mutation Intentional invalid changes detected by validators.

Review AI-only, ambiguous, and provisional items remain visible.

Valid Test Data



Apply mutation



Run validator



Detect expected defect

Mutations include invalid segment type, length, values, sequence, Prompt Code, companion segments, lifecycle, and serialization.

+

EXECUTION EVIDENCE

The framework is executable, not only documentary.

Java Reusable validators and report generators.

JUnit Unit, contract, mutation, and package tests.

Cucumber Segment-wise business-readable workflows.

Reports JSON, Markdown, HTML, and PDF evidence.

Source rule



Test package



Validator run



Coverage evidence



Decision record

The current focused Test Solution JSON audit passed with zero failures and zero errors using only Test Solution output and ATL105 knowledge-base inputs.

+